Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/27/2008
PI Name:	Dulchavsky, Scott A. M.D., Ph.D.		
Project Title:	Ground Portion - Advanced Diagnostic Ultrasound in Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sdulcha1@hfhs.org	**Fax:**	FY 313 916 9445
PI Organization Type:	PUBLIC SERVICE	**Phone:**	313 916 9306
Organization Name:	Henry Ford Health System		
PI Address 1:	Surgery		
PI Address 2:	2799 W. Grand Boulevard, CFP-1		
PI Web Page:			
City:	Detroit	**State:**	MI
Zip Code:	48202-2608	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	11/01/2004	**End Date:**	07/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Moved to Henry Ford Hospital from Wayne St U Hosp early 2006 (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-SMS00001		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The diagnosis of acute health problems in space is problematic due to limited training of the Crew Medical Officer (CMO), human and environmental factors, and changes in anatomy and disease presentation in microgravity. Trauma and other acute clinical conditions may cause serious mission ending health consequences. There is no planned radiological capability aboard the ISS further complicating medical diagnosis in space, however, terrestrial investigations have suggested expanded diagnostic applications of ultrasound in abdominal/thoracic trauma, bony and ligamentous injury, dental/sinus infections, opthalmic, and urologic complications. The Health Research Facility (HRF) will have ultrasound capability for research and possibly operational use. This proposal will optimize diagnostic ultrasound in these applications on the ground to fulfill the objectives of this proposal: Maximize the sensitivity and specificity of ultrasound in the diagnosis of abdominal injury, pneumothorax, extremity injury, dental/sinus infection, opthalmic injury, and urologic disorders in ground based testing.		

	The unique constraints imposed by the space environment require the development of novel strategies for crew member health problems including the expanded use of ultrasound. Thoracic ultrasound, initially investigated by NASA as an alternative diagnostic modality for pneumothorax and orthopedic injury, has proven accuracy in terrestrial and microgravity applications and may have wide spread impact in acute care. The ultrasonic diagnostic modalities described in this proposal, when optimized, would provide a significant, clinically relevant advance in space medical capabilities with profound Earth-based ramifications in rural or military applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ultrasonic diagnostic modalities described in this proposal, when optimized, would provide a significant, clinically relevant advance in space medical capabilities with profound Earth-based ramifications in rural or military applications.
Task Progress:	We have optimized numerous non-traditional ultrasound techniques including musculoskeletal, ocular, dental, and genito-urinary ultrasound. We have also determined the human factor requirements for performing ultrasound examinations in microgravity using the NASA microgravity research facility. We have verified the accuracy of ultrasound in non-traditional applications through ground based investigations.
Bibliography Type:	Description: (Last Updated: 07/24/2025)
Articles in Peer-reviewed Journals	Fincke EM, Padalka G, Lee D, van Holsbeeck M, Sargsyan AE, Hamilton DR, Martin D, Melton SL, McFarlin K, Dulchavsky SA. "Evaluation of shoulder integrity in space: first report of musculoskeletal US on the International Space Station." Radiology. 2005 Feb;234(2):319-22. PMID: 15533948 , Feb-2005
Articles in Peer-reviewed Journals	McFarlin K, Sargsyan AE, Melton S, Hamilton DR, Dulchavsky SA. "A surgeon's guide to the universe." Surgery. 2006 May;139(5):587-90. PMID: 16701089 , May-2006
Articles in Peer-reviewed Journals	Dulchavsky SA, Sargsyan AE, Garcia KM, Melton SL, Ebert D, Hamilton DR. "Intuitive ultrasonography for autonomous medical care in limited-resource environments." Acta Astronaut. 2011 May-Jun;68(9-10):1595-607. http://dx.doi.org/10.1016/j.actaastro.2009.08.024 , May-2011